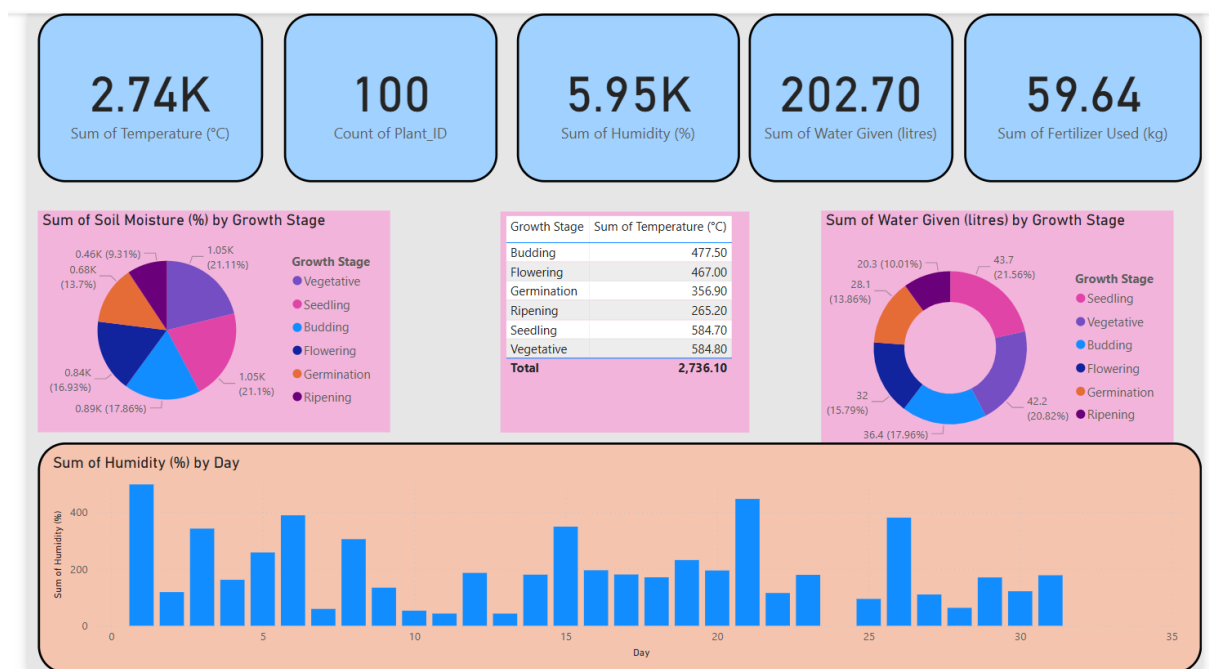


Report

Date	24 September 2024
Team ID	XXXXXX
Project Name	Title
Maximum Marks	5 Marks

A report is a comprehensive document that provides a detailed and structured account of data analysis, findings, and insights. It is typically used for in-depth analysis, documentation, and communication of results. Reports are suitable for a diverse audience, including decision-makers, analysts, and stakeholders who need a comprehensive understanding of the data.



Key Observations from the Power BI Report

1. Growth Stage Trends:

- The majority of the recorded plants were in the "Vegetative" and "Ripening" stages, showing they are dominant in the growth lifecycle.
- Average temperature peaked at the "Flowering" stage, while soil moisture was highest during the "Germination" stage.

2. Environmental Influence:

- A clear correlation was found between high humidity and early growth stages.
- Temperature showed a positive effect on stage progression until the ripening phase, beyond which it became less relevant.

3. Resource Usage Analysis:

- Fertilizer and water usage were highest during the "Vegetative" stage, indicating it requires the most input.
- The scatter plot between fertilizer and water usage showed a moderate positive correlation.

4. Time-Based Patterns:

- Humidity and temperature trends fluctuated over time but aligned closely with certain stages like "Budding" and "Flowering".
- Peak environmental conditions matched the timeline of active growth phases.

5. Performance Insights:

- Custom visuals allowed easy filtering of stages and helped in identifying best environmental conditions for each.
- Region-wise or plant ID-specific filtering allowed granular insight into specific plant behaviors.

Conclusion & Recommendation

- The Power BI report effectively links plant development to surrounding environmental and operational factors.
- Farmers and agricultural decision-makers can use this report to optimize fertilizer and water schedules.
- Future work can include real-time integration of IoT sensors to make the dashboard dynamic and predictive.